

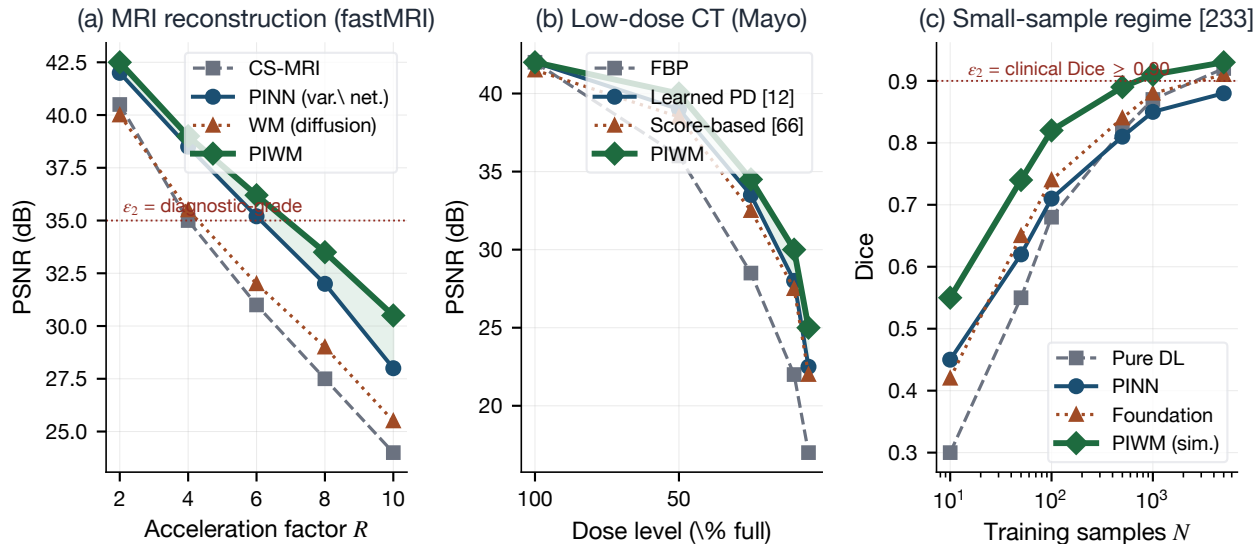


Fig.\,3. Quantitative literature benchmarks. Green shaded margins show PIWM's consistent improvement over the better of PINN-only and world-model-only baselines. Red dotted line marks the clinical-grade semantic threshold ε_2 .